Debayan Lahiry

Cybersecurity Student
+91 9321128211 | debayan.lahiry@gmail.com | Mumbai, India
LinkedIn | GitHub | Portfolio

PROFESSIONAL SUMMARY

Cybersecurity undergraduate (B.Tech CSE, 2028) building security systems: a SIEM-inspired detection platform, an ML prompt-injection detector and an AWS serverless platform. Founder of InboxIQ, a deployed B2B SaaS.

EDUCATION

B.Tech, Computer Science and Engineering (Cybersecurity) Expected 2028
Mukesh Patel School of Technology Management & Engineering (MPSTME), NMIMS University, Mumbai

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, SQL

Security: Threat Detection, Event Correlation, MITRE ATT&CK, Application Security, OAuth 2.0, RBAC, Prompt-Injection Detection, Cloud Security (IAM, encryption)

Backend & Databases: FastAPI, Node.js, Express, REST APIs, PostgreSQL, Supabase, SQLAlchemy, Alembic

Cloud & DevOps: AWS Lambda, API Gateway, S3, DynamoDB, SNS, IAM, AWS SAM, Git, GitHub Actions, pytest

Frontend: React, Next.js, Vite, Tailwind CSS

ML & Data: scikit-learn, pandas, NumPy, TF-IDF, Logistic Regression

SELECTED PROJECTS

AEGIS-X | SIEM-Inspired Security Monitoring Platform GitHub
Python, FastAPI, SQLAlchemy, Alembic, PostgreSQL/SQLite, Next.js, React, TypeScript, GitHub Actions

- Engineered an event-ingestion API and eight deterministic five-minute-window detection rules (brute force, login after repeated failures, port scans, encoded commands), mapped to MITRE ATT&CK and correlated into evidence-linked incidents.
- Implemented Argon2 hashing, revocable HTTP-only sessions, analyst/admin access control, origin checks and rate limiting; tested on synthetic telemetry (23 passing pytest tests, GitHub Actions CI).

InboxIQ | AI Email Productivity SaaS (Founder & Full-Stack Developer) Live SaaS | GitHub
React, TypeScript, Vite, Tailwind CSS, Node.js, Express, PostgreSQL (Supabase), BetterAuth, Gmail API, Groq LLM API

- Architected and deployed a full-stack B2B SaaS for email productivity, lead detection and follow-ups: React/TypeScript frontend on Vercel, Node.js/Express API, Supabase PostgreSQL, Gmail API and LLM-assisted classification and drafting.
- Implemented BetterAuth and Gmail OAuth (HMAC-signed, expiring, user-bound state) with per-user query scoping and capped mailbox imports.

InjeX | Prompt Injection Detection & Adversarial Robustness Testing Live frontend
Python, scikit-learn, pandas, React, TypeScript

- Built a prompt-injection classifier (word- and character-level TF-IDF with Logistic Regression) trained on a public prompt-injection dataset, with Unicode normalisation and zero-width-character removal in preprocessing.
- Generated adversarial prompt variants (character substitution, spacing, punctuation, instruction splitting, Unicode obfuscation) to train an augmented detector, then compared it with a baseline on accuracy, precision, recall, F1 and false-positive rate.

DrainSentinel | AWS Serverless Storm-Drain Risk Platform GitHub
Python, FastAPI, AWS Lambda, API Gateway, DynamoDB, S3, SNS, AWS SAM, React, TypeScript, Leaflet

- Built a FastAPI backend for AWS Lambda (Mangum) behind API Gateway, with DynamoDB, S3 and SNS provisioned through an AWS SAM template, and a React/TypeScript/Leaflet dashboard mapping drain risk and a prioritised inspection queue.
- Applied cloud security controls in the template: scoped per-table IAM policies, DynamoDB encryption at rest with point-in-time recovery, a private S3 bucket (Block Public Access, default encryption, presigned URLs), X-Ray tracing and a NoEcho secret parameter; 33 passing pytest tests.

CERTIFICATIONS & TRAINING

- **AWS Academy Graduate — Cloud Security Foundations** (AWS training badge) · **Introduction to Cloud Security** (Simplilearn SkillUp)
- **Advanced Malware and Network Anomaly Detection** (Johns Hopkins University / Coursera) · **Network Defense** (Cisco Networking Academy)
- **Operationalizing Cloud Security with Deepfence ThreatMapper** (Udacity) · **Google Cloud Computing Foundations** (Google Cloud)
- **Introduction to Microsoft Purview and Microsoft's Privacy Principles** (Microsoft)
- **AWS Training & Certification modules (not AWS certifications):** GuardDuty, Detective, Inspector, Security Lake, Secrets Manager, Network Firewall, Firewall Manager, Incident Response, Encryption, IAM, Cognito, Identity Center.